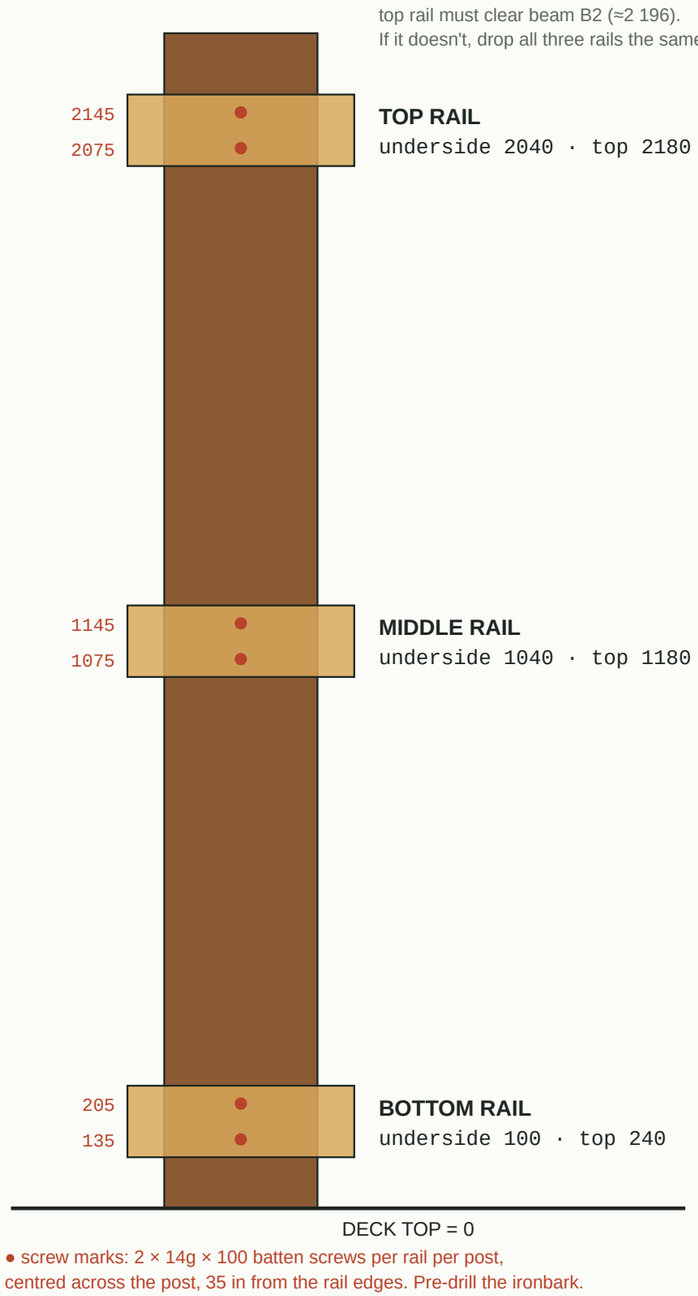
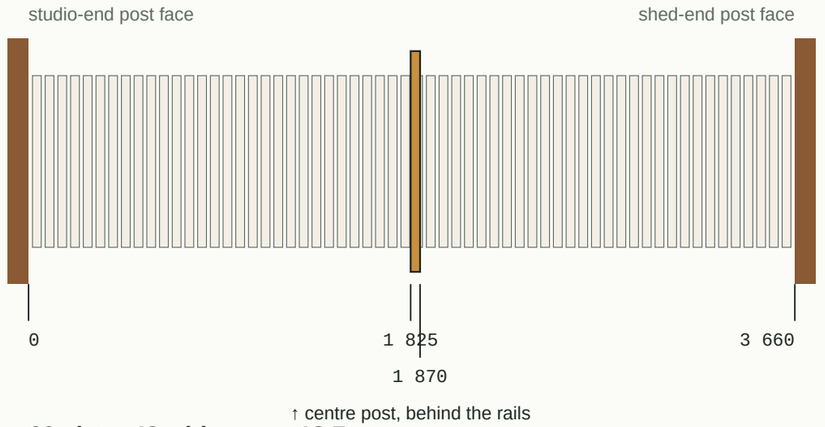
1. MARK BOTH IRONBARK POSTS

Fence-side face. Measure up from the deck top (use a level line, not the rafter).



● screw marks: 2 × 14g × 100 batten screws per rail per post, centred across the post, 35 in from the rail edges. Pre-drill the ironbark.

2. ALONG THE WALL (fence side)



60 slats, 42 wide, gaps 18.7

Spacer: an offcut of slat stood on edge (19), nipped up a hair so slat 60 finishes the same gap off the shed-end post.

Tape-check the LEFT edge of these slats from the studio-end post face:

slat 1 **19**	slat 11 **626**	slat 21 **1232**	slat 31 **1839**
slat 41 **2446**	slat 51 **3053**	slat 60 **3599**	

Slat bottoms sit 20 above the deck (use a 20 packer).
Tops run up behind the rafter: 3 cut lengths, see the cut plan.

3. BUY

42×19 H3 primed pine, 5.4 m	32
140×45 H3 MGP10, 4.2 m (middle rail)	1
14g × 100 hex batten screws	12
M12 bolts 3×200, 2×120, 2×220, 1 × 300 rod, nuts, washers	set
16g × 50 galv finish nails	≈400

Plus the 3 leftover joists: 2 rails and the centre post.

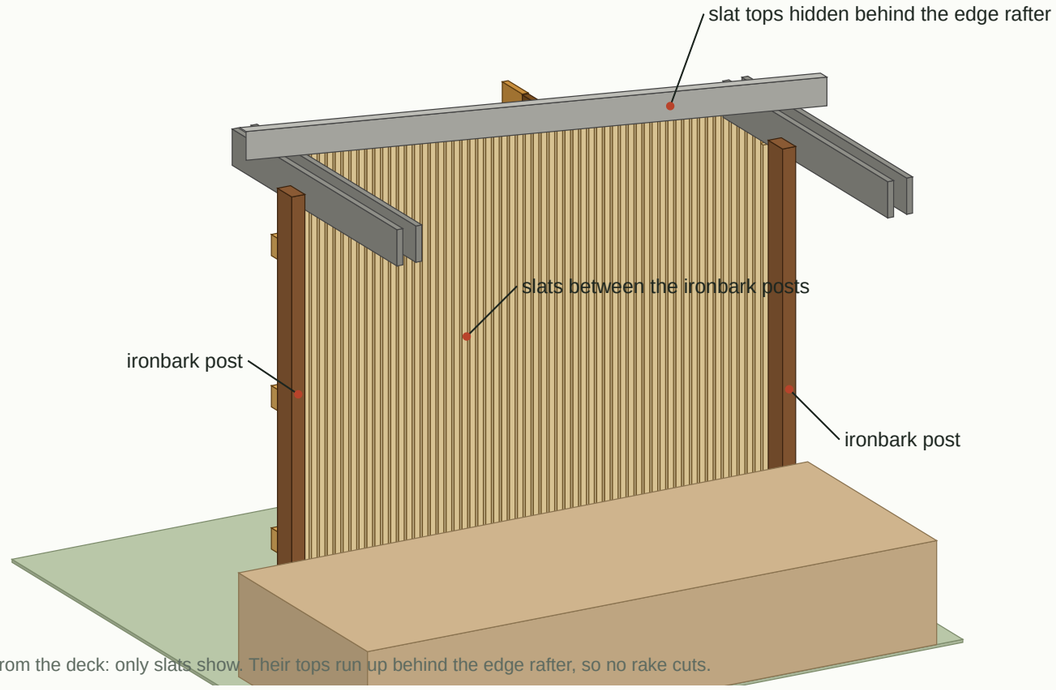
4. CUT THE SLATS

FROM EACH 5.4 M LENGTH	LENGTHS
1 × A (2775) + 1 × C (2565)	20
2 × B (2670)	10
Spare	2
Total 5.4 m lengths	**32**

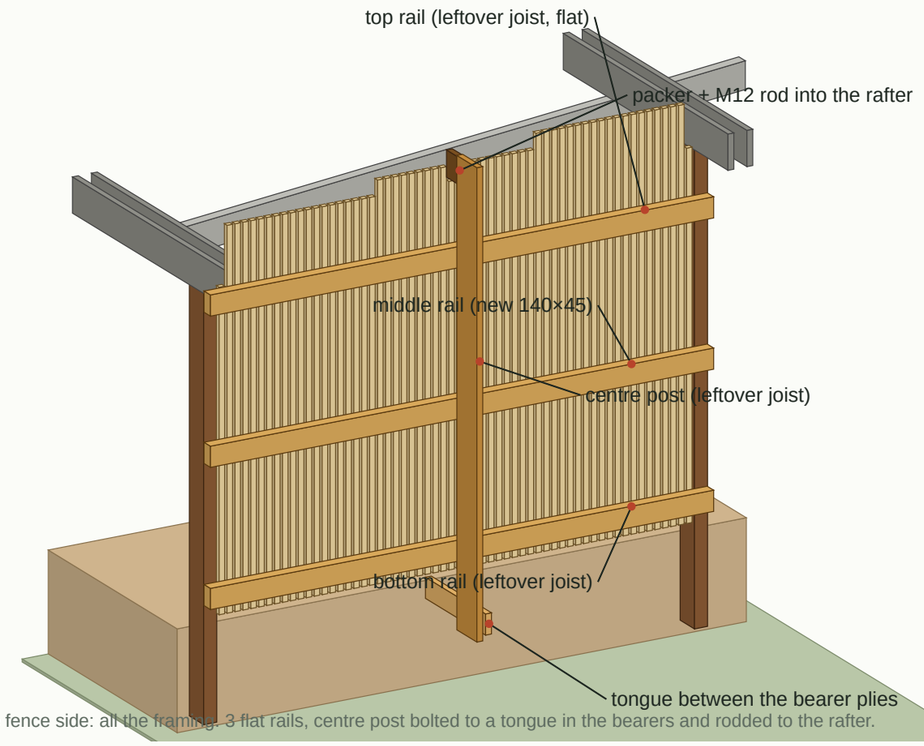
A = slats 1–20 (studio end), B = 21–40, C = 41–60. Three specials come out of the same pieces, cut shorter: slat 1 = 2482, slat 31 = 2569, slat 60 = 2166. Square off the factory ends first; each length leaves about 50 spare.

5. BUILD

1. Mark both ironbark posts (left drawing).
2. Fit the tongue and the centre post, plumb, at 1 825 from the studio-end post face (S4).
3. Screw the 3 rails flat across the posts, bottom first. Bolt each one to the centre post.
4. String-line the rails. Pack at the centre post if needed.
5. Cut the slats (cut plan) and prime the ends.
6. Fix slats from the studio end, 20 off the deck, 2 nails per rail. Tape-check every 10.
7. Topcoat.



From the deck: only slats show. Their tops run up behind the edge rafter, so no rake cuts.



From the fence side: all the framing. 3 flat rails, centre post bolted to a tongue in the bearers and rodded to the rafter.

AT A GLANCE

Clear span between ironbark posts	3 660
Rails (flat, fence face of posts): bottom / middle / top	100–240 / 1 040–1 180 / 2 040–2 180
Centre post, 1 825 from the studio-end post	3136 long
Slats	60 @ 42, gaps 18.7
Slat lengths A / B / C	2775 / 2670 / 2565
Slat bottom above deck	20

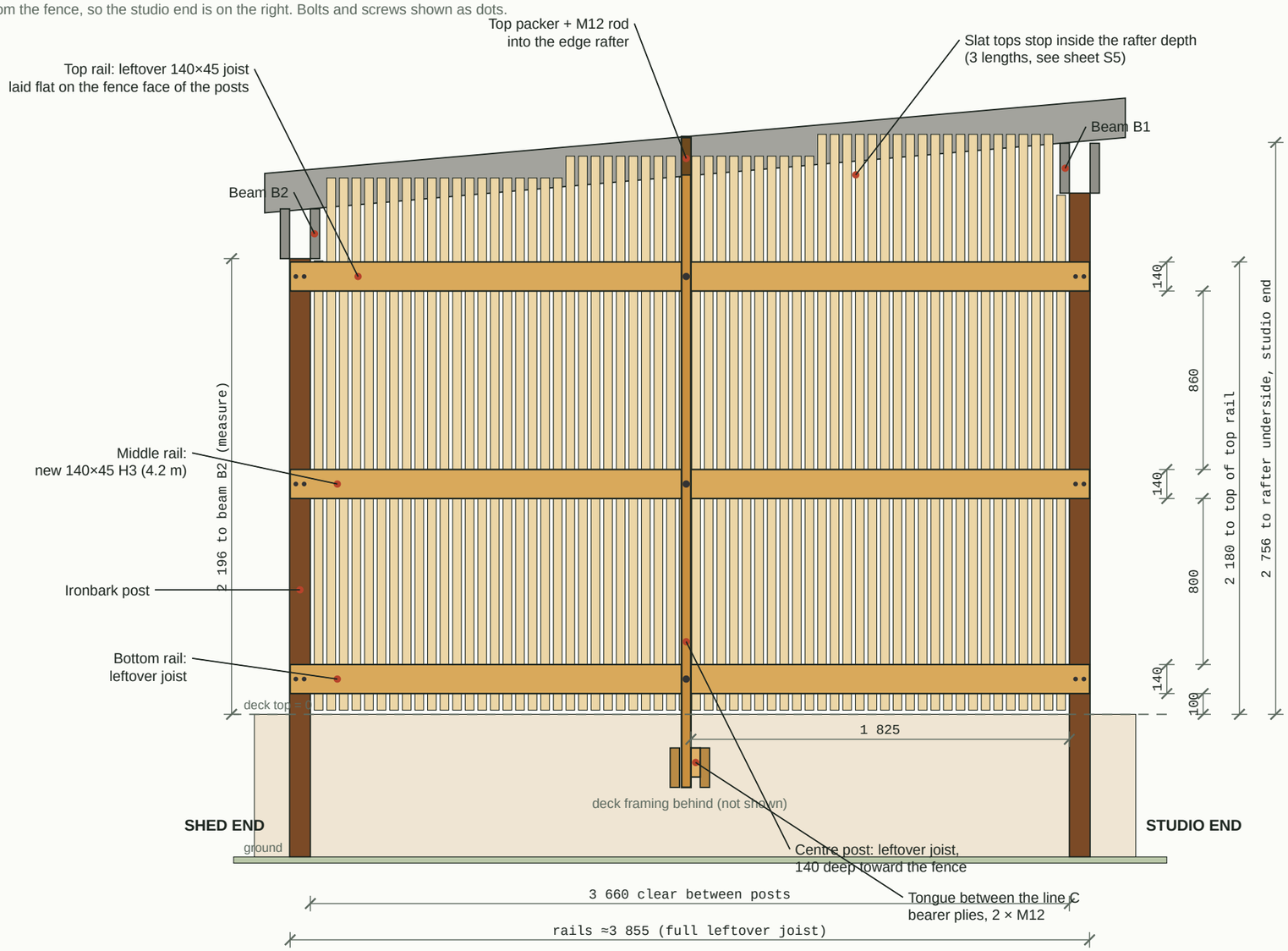
> Why the centre post: a 140×45 rail laid flat can't span the full 3.66 m in a big wind. Halving the span to 1.83 m brings it to about 12 MPa against an MGP10 limit of about 16, with about 8 mm of bow.

MATERIALS AND HARDWARE

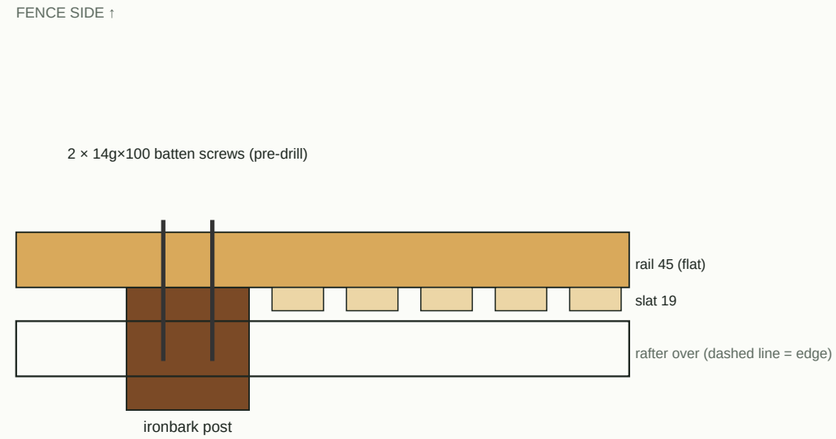
ITEM	SIZE	QTY	USE
Leftover deck joist, H3	140×45 × ≈3 855	3	top rail, bottom rail, centre post (+ 420 tongue from the post offcut)
New H3 MGP10	140×45 × 4.2 m	1	middle rail, trim to match the others
H3 primed pine, 42×19 (Old Mill)	5.4 m	32	60 slats + 2 spare lengths; cut plan on S1. If only 3.0/3.6 m stock: 63 lengths
Offcuts 70×35	≈200	4	top packer (2 flat = 70) and tongue packers
Hex batten screws, Type 17, galv	14g × 100	12	rails to ironbark posts, 2 per crossing, pre-drill
Galv bolts, nuts, 2 washers	M12 × 200	3	each rail to the centre post
Galv bolts, nuts, 2 washers	M12 × 120	2	centre post to tongue
Galv bolts, nuts, 2 washers	M12 × 220	2	tongue through both bearer plies
Galv threaded rod, nuts, washers	M12 × 300	1	post → packer → rafter
Galv or stainless finish nails (nail gun)	16g × 50	≈400	2 per slat per rail. 18g brads are too thin for wind.
Exterior primer + exterior acrylic		1 each	prime every cut end; 2 topcoats on the slats

ELEVATION FROM THE FENCE SIDE (1:25)

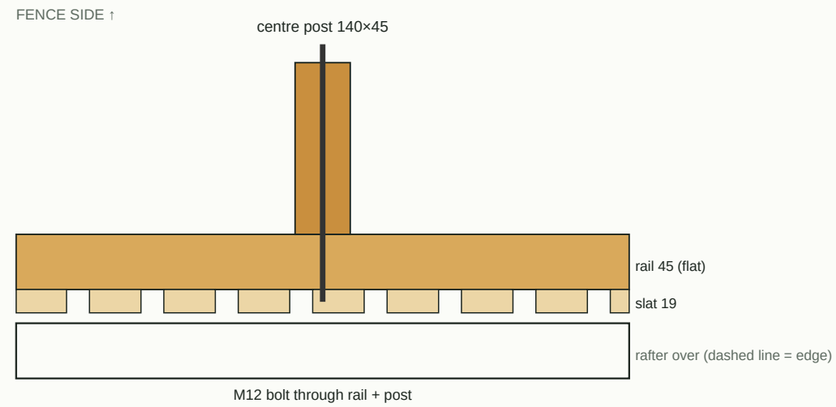
Shown from the fence, so the studio end is on the right. Bolts and screws shown as dots.



PLAN AT A RAIL, IRONBARK POST (1:5)

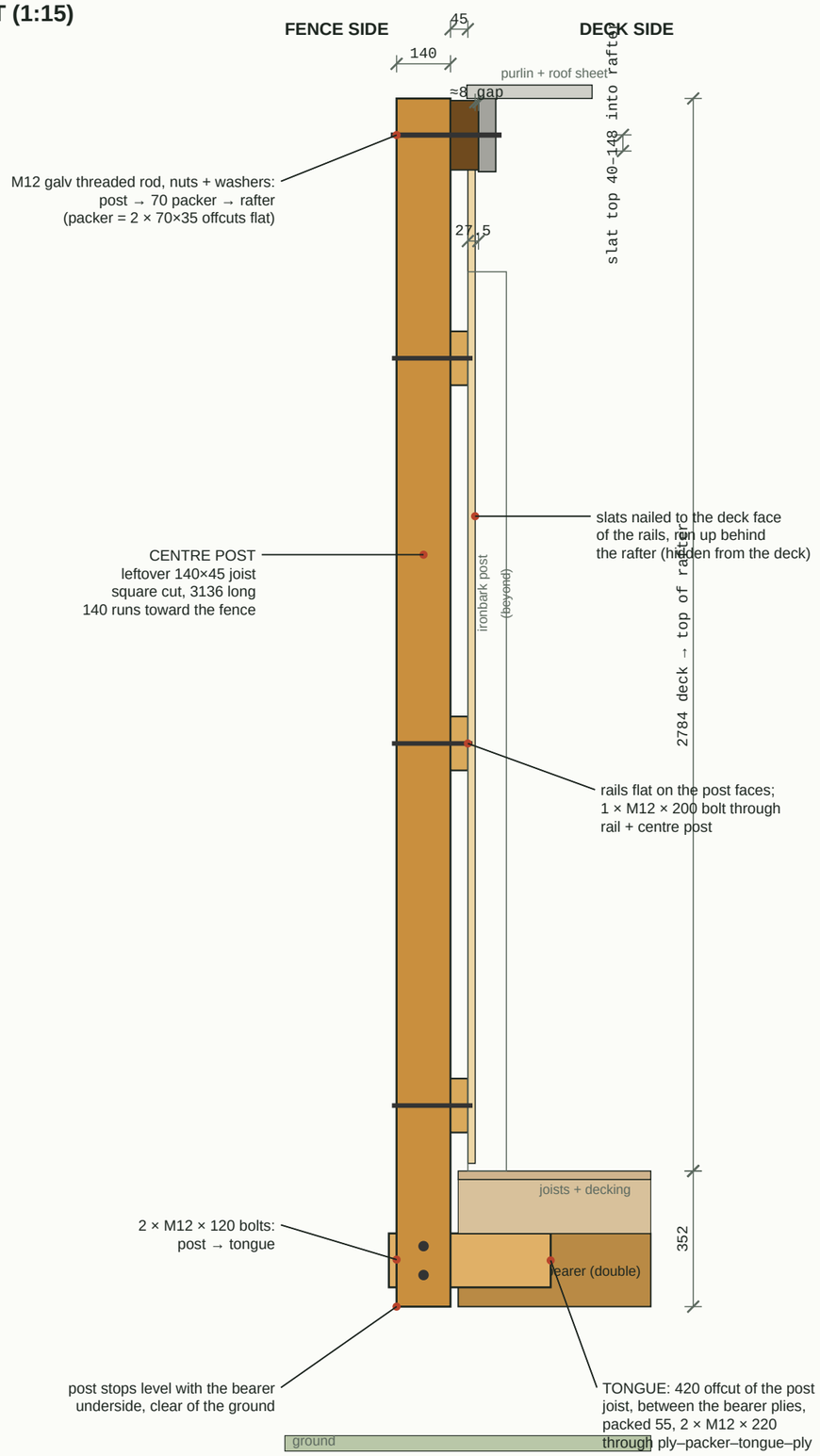


PLAN AT A RAIL, CENTRE POST (1:5)



Check on site before cutting: clear span, rafter underside at both ends, beam B2 underside, slat-to-rafter gap. Structure to be confirmed by Ollie Harris before building.

SECTION THROUGH THE CENTRE POST (1:15)



FIXING SCHEDULE

JOINT	FIXING
Rail → ironbark post	2 × 14g × 100 hex batten screws per crossing, pre-drilled 5 mm (ironbark splits)
Rail → centre post	1 × M12 × 200 bolt per rail, through the 45 rail and 140 post, washers both sides
Centre post → tongue	2 × M12 × 120 bolts, 75 apart vertically
Tongue → line C bearer	2 × M12 × 220 bolts through ply, 55 packer, tongue, ply
Centre post top → rafter	1 × M12 threaded rod through post, 70 packer and rafter, washers and nuts both ends
Slat → rail	2 × 50 stainless or galv nails (or 8g × 40 stainless screws) per rail, 15 in from the edges

HOW THE LOADS GO

- Wind on the slats goes into the 3 rails.
- Each rail spans 1.83 m, from an ironbark post to the centre post and on to the other ironbark post.
- The centre post spans about 2.8 m, from the tongue in the bearers up to the rafter, and hands its load to both.
- Slat tops are unfixed and hang up to 615 above the top rail. That's fine for 42 × 19 in strength and stiffness.

The ≈8 mm gap between the slat face and the side of the rafter is from the model. Hold a slat up and check it before you cut. If a slat rubs, flip it or take a plane pass off the back.

Optional: to stop the slat tops rattling, put one 8g screw through each slat into the side of the rafter, with an 8 mm packer strip behind. Screw from the fence side so it can't be seen from the deck.

SLAT CUTTING SCHEDULE

GROUP	SLAT NO. (FROM THE STUDIO END)	CUT LENGTH	TOP ABOVE DECK	NOTE
A	1–20	2775	2795	studio end → shed end: tops sit 41–145 into the rafter
B	21–40	2670	2690	studio end → shed end: tops sit 42–146 into the rafter
C	41–60	2565	2585	studio end → shed end: tops sit 43–147 into the rafter
—	1	2482	2502	stops 10 under beam B1
—	31	2569	2589	stops 10 under the top packer
—	60	2166	2186	stops 10 under beam B2

The rafter drops about 316 across the span, so one length can't stay hidden behind it the whole way. The three groups each step down 105, which keeps every top 40–148 up inside the 190 rafter: hidden from the deck and clear of the purlins.

CHECKS BEFORE YOU CALL IT DONE

- No slat top shows below the rafter from a seated position on the deck.
- No slat touches the purlins or the roof sheet.
- Rails don't touch beam B2 (top rail 2 180 vs beam ≈2 196).
- Centre post plumb, nuts tight, rod snug.
- All cut ends sealed.

> Wind assumption: about 1.1 kPa ultimate on a screen that's 80% solid, with MGP10 H3 framing. The joints at the tongue and the rafter take the most load. Ollie to confirm before building.

BUILD ORDER

1. **Measure first.** Check the clear span between the posts (3 660), the rafter underside at both ends (≈2 756 studio, ≈2 436 shed, above the deck), beam B2's underside (≈2 196) and the slat-to-rafter gap. If any figure is more than 20 off, adjust the slat lengths in the table.
2. **Mark the rail heights** on the fence face of both ironbark posts: 100, 1 040 and 2 040 to the underside of each rail. Use a level or laser off the deck, not the rafter.
3. **Cut the centre post joist:** post 3136 square both ends, and the 420 tongue from the offcut. Seal the ends.
4. **Fit the tongue.** Slide it between the two line C bearer plies at the fence-side end, against the studio-side ply. Pack the 55 gap with 70×35 offcuts, then bolt with 2 × M12 × 220. It projects 180 past the bearer ends.
5. **Stand the centre post.** Bolt it to the fence-side end of the tongue with its deck-side face exactly one rail thickness (45) back from the fence face of the ironbark posts, so the rails sit flat across all three. Clamp it, plumb it both ways, then drill and fit the 2 × M12 × 120.
6. **Fix the top.** Pack between the post and the rafter with 2 × 70×35 offcuts laid flat (70). Drill through post, packer and rafter and fit the M12 rod. Snug it, don't crush it.
7. **Hang the rails.** Bottom first, then middle, then top. Clamp each one on its marks, check it's level, pre-drill and fix 2 batten screws into each ironbark post, then drill and bolt through the centre post.
8. **String-line check:** the deck face of the 3 rails should be dead straight. Pack behind at the centre post if needed before the slats go on.
9. **Cut the slats** to the 3 lengths, plus the 3 specials (table). Seal the cut ends. Each length is one stop-block setting.
10. **Slat set-out:** start at the studio-end post with a 18.7 mm gap (an offcut stood on edge, nipped up slightly), bottoms on a 20 packer off the deck. Nail 2 per rail (16g × 50). Check plumb every 10 slats against the tape table on S1 and fudge the gaps slightly if needed so the last gap at the shed-end post matches.
11. **Finish:** prime all cut ends, then 2 topcoats of exterior acrylic. Primer alone isn't weatherproof.